MIL-I-38120A(USAF)
8 March 1963
Superseding
MIL-I-38120 (USAF)
2 November 1962

MILITARY SPECIFICATION

INDICATOR, RATE OF FLOW AND TRANSFER EFU-20/A

1. SCOPE

1.1 This specification covers one type of in-flight refueling rate of flow and fuel transfer indicator, designated EFU-20/A.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-636	Box, Fiberboard

Military

MIL-P-116	Preservation, Methods of
MIL-C-5015	Connectors, Electric, AN Type
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation for Delivery
MIL-P-8622	Power Supply, 28V DC Input, 15V 4 Cycle 3 Phase Output, Type MC-1
MIL-T-9107	Test Reports, Preparation of
MIL-L-25142	Luminescent Material, Fluorescent
MIL-T-25439	Transmitter, Rate of Flow, In-Flight Refueling, 0-10,000 PPH, Type MS-3
MIL-I-26600	Interference Control Requirements, Aeronautical Equipment
MIL-D-70327	Drawings, Engineering and Associated Lists

FSC 6680

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:55Z
Check the source to verify that this is the current version before use.

STANDARD

Military

MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS33586	Metals, Definition of Dissimilar
MS33639	Cases, Instrument, Clamp-Mounted, Aircraft
MS33678	Connector, Receptacle, Electric, Integral Mounting

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 QUALIFICATION.- The indicator furnished under this specification shall be a product which has been tested and passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products list.

3.2 SELECTION OF SPECIFICATIONS AND STANDARDS.- Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 PRELIMINARY LIST OF ELECTRONIC PARTS.- A preliminary list of electronic parts shall be furnished directly to the procuring activity in accordance with MIL-E-5400 prior to ordering or fabricating parts for the production quantity. The parts shall conform to MIL-E-5400. Nonstandard parts shall be approved by the procuring activity prior to the fabrication of the completed item. Whenever practicable, standard electronic parts shall be used.

3.4 MATERIALS

3.4.1 NONMAGNETIC MATERIALS.- Nonmagnetic materials shall be used for all parts of the indicator, except where magnetic materials are essential.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:55Z
Check the source to verify that this is the current version before use.

sensitive tape.

3.4.2 NONFERROUS MATERIALS.- Nonferrous materials shall be used for all parts of the indicator, except where ferrous materials are essential.

3.4.3 METALS.- Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service.

3.4.4 DISSIMILAR METALS.- Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

3.4.5 TOXIC AND CORROSIVE FUMES.- The indicator shall be of such material and construction that, when the material is burned, fumes or gases that combine with the atmosphere to form acid or corrosive alkali will not be liberated.

3.4.6 PROTECTIVE TREATMENT.- When materials are used in the construction of the indicator that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.5 DESIGN AND CONSTRUCTION

3.5.1 FUNCTION.- The indicator shall be designed as a combined amplifier-indicator to indicate the rate of flow in pounds per minute (ppm) by means of an index-type pointer moving around the periphery of the dial from 400 to 10,000 ppm. Fuel consumed from 0 to 999,000 pounds shall be indicated by a cyclometer located in the central part of the dial. When connected in conjunction with a type MB-3 transmitter conforming to MIL-T-25439 and its associated power supply in accordance with MIL-P-8422, the indicator shall operate with an input voltage of 1V per 1,000 ppm fuel flow rate.

3.5.2 POWER SUPPLY.- The indicator shall be designed to operate from a 115V, 400 cycle, single-phase power supply in accordance with MIL-STD-704, except that the voltage and frequency may vary together or independently ±5 percent.

3.5.2.1 POWER CONSUMPTION.- The total power consumption of the indicator shall not exceed 30 va.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:55Z
Check the source to verify that this is the current version before use.

3.5.3 ANTICREEP PROVISIONS.- The indicator shall incorporate anti-creep provisions for low flow rates which shall exhibit the following characteristics:

a. As the fuel flow rate input signal increases from zero, the counter servo shall be energized when the fuel flow rate is between 200 and 290 ppm

b. As the fuel flow rate decreases from a value above 290 ppm, the counter servo shall become deenergized when the flow rate is between 180 and 125 ppm.

3.5.4 CASE.- The case shall conform to MS33639, shall be finished in durable dull black in accordance with color No. 37038 of FED-STD-595, and shall be hermetically sealed. The seal shall be so designed and constructed that it will not be dependent upon materials that might be affected by the action of any atmospheric conditions to which the case may be subjected. The front edge of the case shall not extend beyond the outer surface of the cover glass by more than 1/32 inch. The case length shall not exceed 7-13/16 inches, exclusive of the electrical receptacle. The design of the case shall be such that the internal mechanism may be removed from the case, replaced, and the case resealed without the use of special tools, unless approved by the procuring activity.

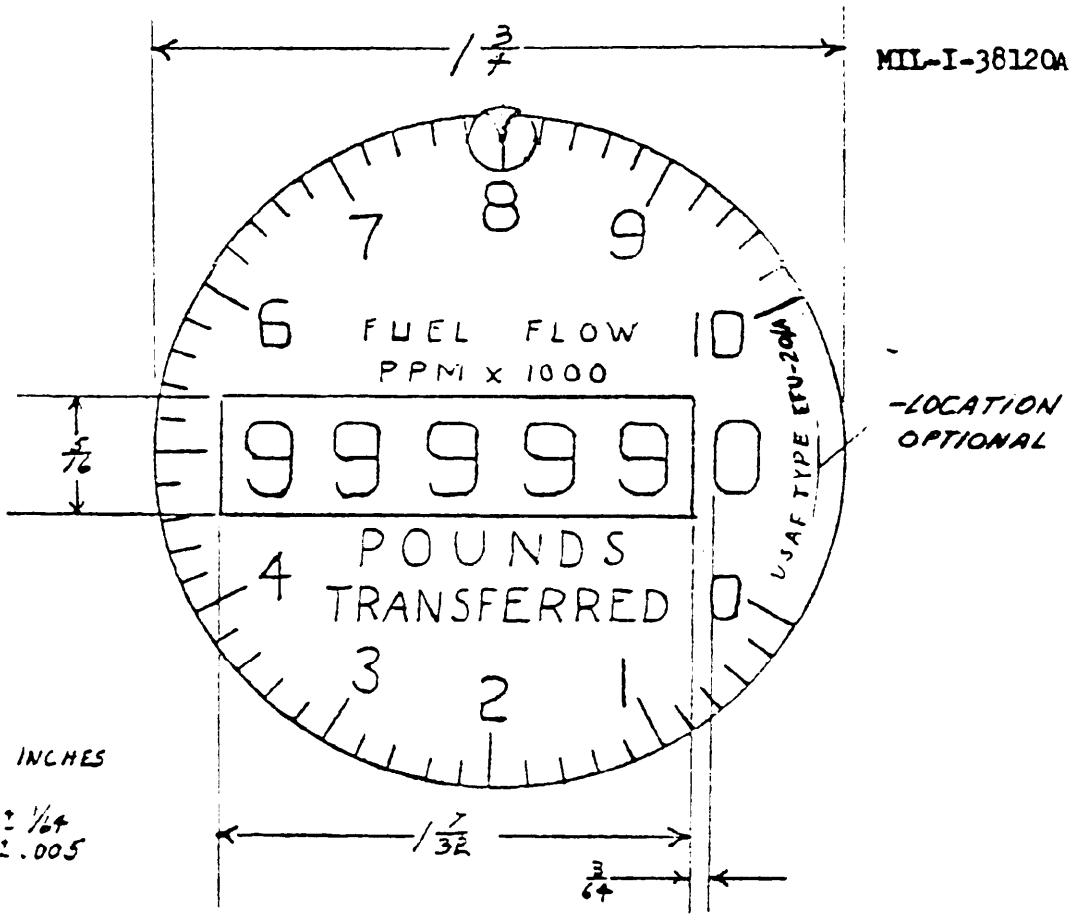
3.5.4.1 FILLING MEDIUM.- The filling medium shall be of at least 98 percent purity, free of dust particles, and shall contain not more than 0.006 milligram of water vapor per liter (dew point -65° C) at the filling pressure. The filling medium shall be either 100 percent helium or a mixture of 88 to 92 percent nitrogen and the remainder helium. The absolute pressure of the filling medium in the case shall be approximately 1 or 1/2 atmosphere, as required. Where practicable, the filling medium shall be 100 percent helium.

3.5.4.2 FILLER TUBE.- A tube or other facility for evacuating and filling the case shall be provided at the rear of the case and shall be recessed beneath the level of the back plate surface.

3.5.4.3 DIAL.- The indicator dial shall conform to figure 1. The style and proportions of numerals and letters placed on the dial shall conform to MS33558 except for the counter numerals. Numerals shall distinctly indicate the graduations to which they apply.

3.5.4.3.1 DIAL, POINTER, AND COUNTER MARKINGS.- The dial, pointer, and counter markings shall be finished in fluorescent-luminescent material in accordance with MIL-L-25142, type I or III, as applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:55Z
Check the source to verify that this is the current version before use.



	Height or Length Inch ±1/64	Width of Line or Graduation Inch ±0.005
Scale numerals 0, 1, 2, 3, 4, 6, 7, 8, 9, and 10	5/32	0.020
Major Graduations	5/32	0.020
200 Pound graduations	3/32	0.016
Shaded portion of index pointer		
Lettering "FUEL FLOW" "PPM by 1000"	1/16	0.016
Lettering "POUNDS TRANSFERRED"	3/32	0.020
Counter numerals	3/16 by 1/8	

FIGURE 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:55Z
Check the source to verify that this is the current version before use.

3.5.4.3.1.1 DIAL MARKINGS.- The following markings shall be permanently and legibly marked on the dial. These and all other markings not otherwise specified shall be finished in durable dull black. The height of the letters in USAF TYPE EFU-20/A shall be 3/64 inch.

3.5.4.4 INDEX POINTER.- The length of the index pointer shall not overlap more than two-thirds of the shortest graduation, and the support ring, if used, shall not exceed the length of the largest graduation nor overlap any of the scale numbers.

3.5.4.5 COVER GLASS.- The cover glass shall be clear, flat, and free from flaws which interfere with the normal reading of the instrument. The glass shall not be fluorescent when exposed to ultraviolet light or rays in the 365 millimicron region.

3.5.4.5.1 DIAL TO COVER GLASS DISTANCE.- The maximum distance between the inner surface of the cover glass and the indicator dial shall be 0.188 inch.

3.5.4.6 RATE POINTER STOPS AND CLUTCHES.- Clutches or pointer stops shall be used to limit pointer travel within 1/8 inch beyond each end of the scale.

3.5.5 ELECTRICAL RECEPTACLE.- The indicator shall be provided with an MS33678-18S-1P hermetic-seal receptacle in accordance with MIL-C-5015.

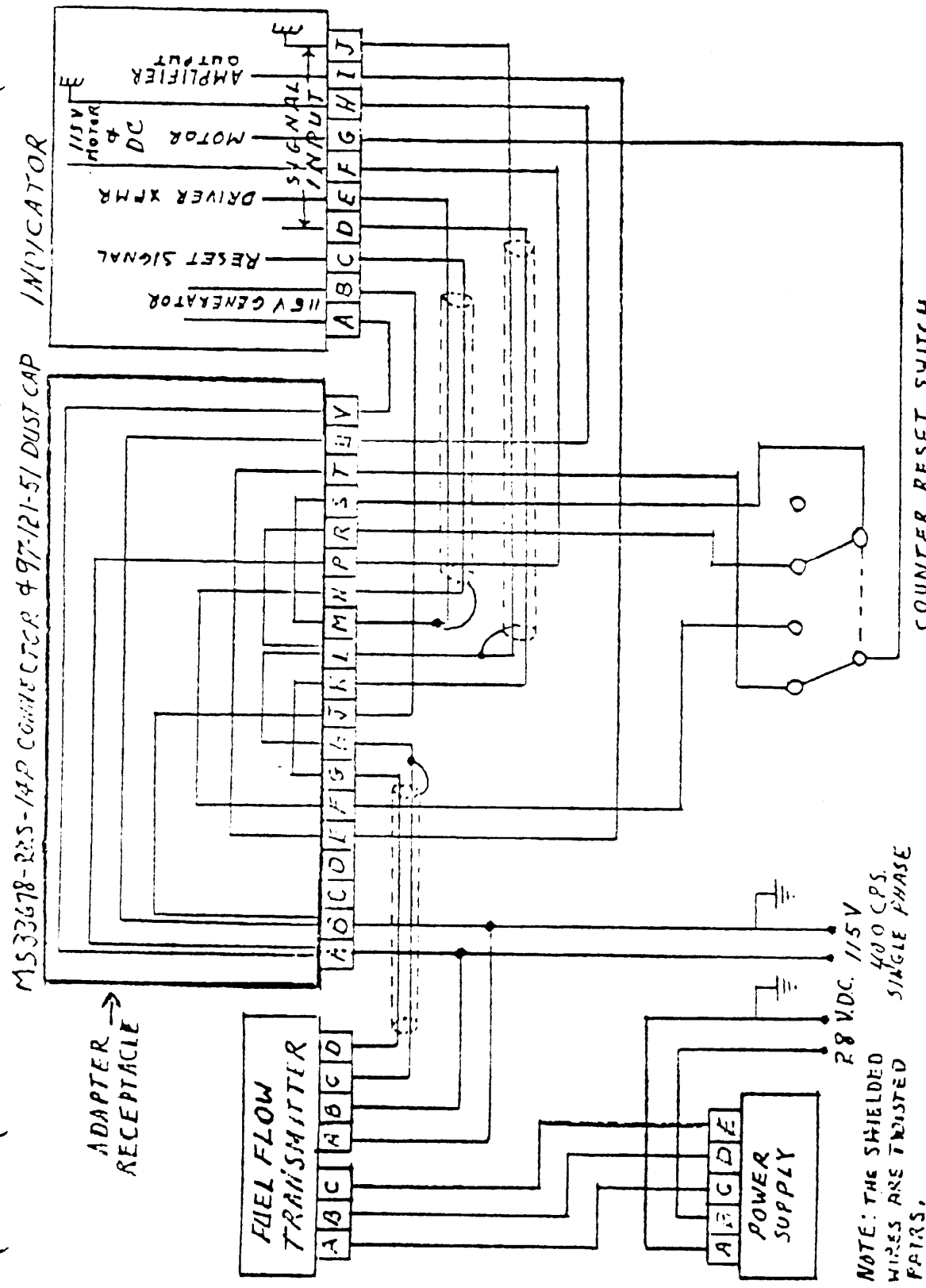
3.5.5.1 ADAPTER RECEPTACLE.- An adapter electrical receptacle, wired as shown on figure 2, shall be furnished by the indicator manufacturer with each indicator. This adapter shall be used in lieu of the amplifier component on existing installations. When the combined indicator-amplifier package specified herein is used in the system, the external wiring diagram shown on figure 2 shall be used for existing in-flight refueling systems.

3.5.5.2 NEW INSTALLATIONS.- New aircraft instrument systems shall not require the adapter receptacle and shall be wired in accordance with figure 3.

3.5.6 TRANSISTORS AND DIODES.- Transistors and diodes used in the construction of the indicator shall be of the silicon type, and for continuous operation in accordance with MIL-STD-70L at a 70° C ambient temperature. Circuit design shall be such that no electronic component will be operated outside its rated characteristics with 115V ±10 percent 400 cps ±10 percent power input and with indicator input signals as specified herein.

3.5.7 REGULATED VOLTAGES.- Regulated voltages, if required, shall be obtained from controlled sources within the indicator. Batteries or standard cells shall not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:55Z
Check the source to verify that this is the current version before use.



ADAPTIVE EXTERNAL WIRING USING EXISTING AIRCRAFT CABLEING

FIGURE 2

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:55Z
Check the source to verify that this is the current version before use.

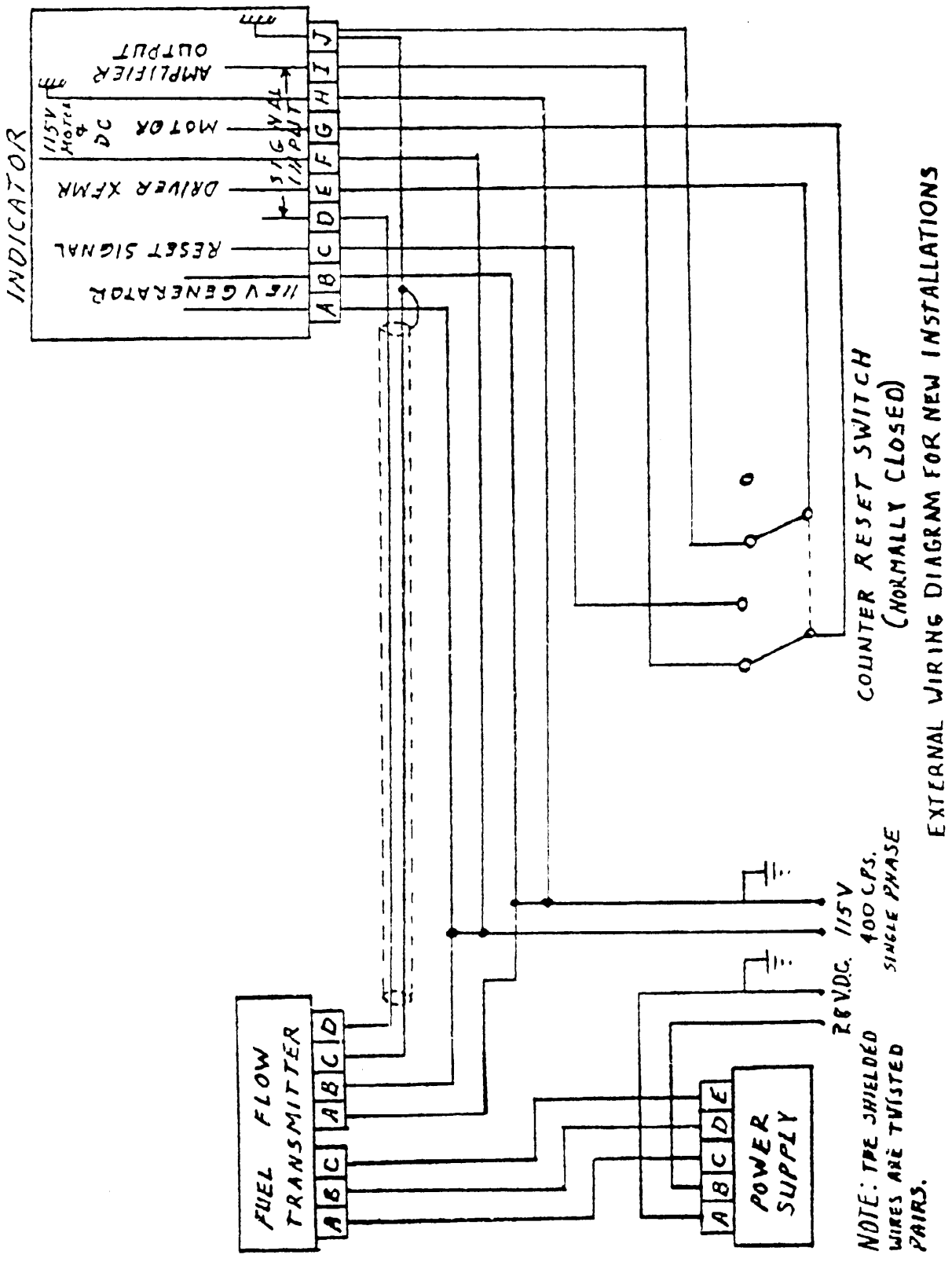


FIGURE 3

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:55Z
Check the source to verify that this is the current version before use.

3.5.8 RESET FACILITIES.- Reset shall be accomplished electrically by reversing the direction of the motor and causing the counter to move all numerals to zero. This may be accomplished by means of the external switch shown on figures 2 and 3.

3.5.9 TEMPERATURE COMPENSATION.- The indicator shall be automatically and internally temperature compensated for the range of -54° to +71° C.

3.6 PERFORMANCE.- The indicator shall be capable of meeting the requirements specified herein when subjected to the following conditions:

a. Temperature - Temperatures within the range of -54 to +71° C

b. Vibration - Vibration for 3 hours at a double amplitude of 0.018 to 0.020 inch and at 500 to 3,000 cpm as specified

c. Salt spray - Exposure to salt sea atmosphere for a 50-hour period

d. Thermal shock - Alternate immersion for 30 minutes in water at +85° C and at +5° C for 8 cycles as specified in section 4.

3.6.1 RADIO NOISE.- Internally radiated and conducted radio noise shall be within the limits specified in MIL-I-26600.

3.6.2 ACCURACY.- Scale error and friction of rate and total consumed indications shall be within the tolerances specified in tables I, II, and III.

TABLE I

Scale Error and Friction						
Test Point PPM	Tolerances					
	Room Temperature		+71° C		-54° C	
	Scale Error ±	Friction	Scale Error ±	Friction	Scale Error ±	Friction
500	90	40	120	40	150	60
1,000	50	40	70	40	85	60
2,000	40	40	55	40	70	60
3,000	60	40	80	40	100	60
4,000	80	40	110	40	140	60
5,000	100	40	130	40	170	60
6,000	120	40	160	40	200	60
7,000	140	40	190	40	230	60
8,000	160	40	210	40	270	60
9,000	180	40	240	40	300	60
10,000	200	40	270	40	330	60

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:55Z
Check the source to verify that this is the current version before use.

3.6.3 ENDURANCE.- The indicator shall be capable of withstanding 1,200 hours of operation as specified.

TABLE II

Counter Error				
Rate of Flow Setting PPM	Test Interval Seconds	Mass & Tolerance (Lbs)		
		Room Temp	High Temp	Low Temp
1,000	120	2,000 ±50	2,000 ±70	2,000 ±90
3,000	120	6,000 ±100	6,000 ±140	6,000 ±180
6,000*	120	12,000 ±150	12,000 ±240	12,000 ±270
9,000	120	18,000 ±300	18,000 ±420	18,000 ±540

TABLE III

Test Point	Test Interval	Mass Tolerance	Scale Error	Friction
6,000 PPM	120 Seconds	12,000 ±300 Lbs	200 PPM	40 PPM
Vibration				
Pointer oscillation 150 ppm Pointer variation 150 ppm				
Voltage and Frequency Variation				
Test Point 3,000 PPM		Tolerance 30 PPM		

3.6.4 RESPONSE.- The indicator shall show a rate indication from zero to full scale or from full scale to zero in less than 2 seconds. Counter response shall be in accordance with table II.

3.7 PART NUMBERING OF INTERCHANGEABLE PARTS.- All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-D-70327 shall govern the manufacturer's part numbers and changes thereto.

3.8 WEIGHT.- The weight of the indicator shall not exceed 2.25 pounds.

3.9 IDENTIFICATION OF PRODUCT.- Equipment, assemblies and parts shall be marked for identification in accordance with MIL-STD-130.

3.9.1 NAMEPLATE.- A nameplate shall be provided on the rear of the case. Where applicable, the information shall conform to MIL-STD-130. Decalcomanias shall not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:55Z
Check the source to verify that this is the current version before use.

3.10 WORKMANSHIP.- The indicator shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, welding, riveting, brazing, plating, painting, machine screw assemblies, and freedom of parts from burrs and sharp edges. Gears shall be of high quality and shall provide a minimum of backlash necessary for smooth operation. Where necessary, antibacklash techniques shall be used.

4. QUALITY ASSURANCE PROVISIONS

4.1 RESPONSIBILITY FOR INSPECTION.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 CLASSIFICATION OF TESTS.- The inspection and testing of indicators shall be classified as follows:

a. Qualification tests

b. Acceptance tests.

4.3 TEST CONDITIONS

4.3.1 STANDARD ATMOSPHERIC CONDITIONS.- Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25° C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.3.2 TAPPING.- Unless otherwise specified, the instrument shall be gently tapped or vibrated before a test reading is taken.

4.3.3 VIBRATION STAND.- Whenever a vibration stand is specified, the stand shall vibrate at any desired frequency between 500 and 3,000 cpm. The stand shall vibrate the components in such a manner that a point on the case will describe, in a plane inclined 45° to the horizontal plane, a circle as specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:55Z
Check the source to verify that this is the current version before use.

4.3.4 TEST POSITION.- Unless otherwise specified, the indicator shall be tested with the mounting surfaces in an upright, vertical position.

4.3.5 MASTER TRANSMITTER.- The indicator shall be tested with a master transmitter especially designed for the purpose. Sufficient tests shall be performed to determine that the transmitted signal is not in error in excess of 0.1 percent. The output voltage of the master transmitter shall have a magnitude of 1V/1,000 ppm at a phase angle of 15° ±0.5° leading the line voltage. Instrument errors due to changes of phase shift in the master transmitter shall be recorded and compensated for in test results.

4.3.6 TOLERANCES.- Test tolerances shall be as specified in tables I, II, and III.

4.4 QUALIFICATION TESTING

4.4.1 TEST SAMPLES.- The qualification test samples shall consist of two indicators representative of the production equipment. The samples shall be identified with the manufacturer's part number and such other information as required by the procuring activity.

4.4.2 TEST REPORT.- When tests are conducted at a location other than the laboratory of the procuring activity, three copies of a test report in accordance with MIL-T-9107 shall be furnished to that activity.

4.4.3 QUALIFICATION TESTS.- The qualification tests shall consist of all tests described under 4.6.

4.5 ACCEPTANCE TESTS.- Acceptance tests shall consist of:

a. Individual tests

b. Sampling tests.

4.5.1 INDIVIDUAL TESTS.- Each indicator shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Dielectric strength

c. Scale error and friction

d. Counter error

e. Electrical characteristics

f. Sealing.

Source: https://assist.dla.mil -- Downloaded: 20260-21T11:55Z
Check the source to verify that this is the current version before use.

4.5.2 SAMPLING PLAN A.- A minimum of 2 indicators shall be selected at random from each 100 or fraction thereof produced and subjected to the following tests:

a. Individual tests

b. Vibration error

c. Vibration endurance

d. Frequency and voltage variation

e. Low temperature

f. High temperature

g. Position error

h. Fogging.

4.5.3 SAMPLING PLAN B.- Unless otherwise specified, 2 indicators shall be selected at random from the first 10 items produced on the contract or order and subjected to the following tests:

a. Sampling plan A tests

b. Low temperature exposure

c. High temperature exposure

d. Salt spray

e. Thermal shock

f. Endurance.

4.5.3.1 REJECTION AND RETEST.- When an item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all tests shall be repeated.

4.5.3.2 INDIVIDUAL TEST MAY CONTINUE.- For production reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or produced later shall not be made until it is determined that all items meet all the requirements of the specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:55Z
Check the source to verify that this is the current version before use.

4.5.4 DEFECTS IN ITEMS ALREADY ACCEPTED.- The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and the method of correcting them.

4.6 TEST METHODS

4.6.1 EXAMINATION OF PRODUCT.- The indicator shall be inspected to determine compliance with this specification with respect to material, workmanship, marking, and dimensions.

4.6.2 DIELECTRIC STRENGTH.- Prior to sealing, 500V at a commercial frequency shall be applied between certain pins and the case for a period of 5 seconds. Only those pins not used for grounding the components to the case shall be used. There shall be no insulation breakdown.

4.6.3 SCALE ERROR AND FRICTION.- The indicator shall be connected to a master transmitter. After the system has operated for 10 minutes, the transmitter shall be so adjusted that when the indicator is at the test point marked with an asterisk in table II, the counter time will be 120 ±0.4 second or 180 ±0.6 second, as applicable. The transmitter reading shall be recorded for use in future inspection tests. The master transmitter shall be set at values equivalent to each of the test points in table I. Two rate-of-flow readings shall be taken, the first before and the second after the instrument is vibrated. The difference between the first and second readings shall not exceed the friction tolerances listed in table I. The second reading shall not exceed the scale error tolerances specified in table I.

4.6.4 COUNTER ERROR.- The fuel flow rates specified in table II shall be set on the master transmitter. The flow indicated on the counter during the test shall be measured and shall be within the tolerances specified in table II.

4.6.5 ELECTRICAL CHARACTERISTICS.- Sufficient electrical tests shall be performed on the indicator and its components to insure satisfactory performance when connected into the flowmeter system. Anticreep tests shall form a part of this test.

4.6.6 SEALING.- The indicator case shall be tested for leaks by means of a mass-spectrometer-type helium leak detector. Where a nitrogen-helium mixture is utilized as the filling medium, the detected leak rate shall not exceed 0.1 micron cubic foot per hour at a pressure differential of 1 atmosphere. Where 100 percent helium is utilized as the filling medium, the leak rate shall not exceed 1.0 micron cubic foot per hour at a pressure differential of 1 atmosphere.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:55Z
Check the source to verify that this is the current version before use.

4.6.7 VIBRATION ERROR.- The indicator shall be subjected to vibration at frequencies within the range of 500 to 3,000 cpm in such a manner that a point of the case will describe, in a plane inclined 45° to the horizontal plane, a circle of approximately 0.008 to 0.011 inch diameter. During this time, the master transmitter shall be set at mid-scale. The pointer oscillation and variation shall not exceed the tolerance specified in table III.

4.6.8 VIBRATION ENDURANCE.- The indicator shall be subjected to vibration for a period of 3 hours on a vibration stand so that a point on the indicator will describe, in a plane inclined 45° to the horizontal, a circle 0.018 to 0.020 inch diameter. During the period of vibration, a master transmitter shall be set at approximately mid-scale. The frequency shall be varied from 500 to 3,000 cpm and the pointer action noted for resonant points. If the indicator does not indicate any resonance within the frequency range, the frequency shall be set at any point between 2,000 and 3,000 cpm. If resonance is noted, the indicator shall vibrate at that point for the duration of the test. Following the vibration period, the indicator shall be checked for scale error and friction. The scale error and friction shall not exceed room temperature tolerances. No screws nor other parts shall become loosened or damaged as a result of vibration.

4.6.9 FREQUENCY AND VOLTAGE VARIATION.- The master transmitter shall be set to the test point listed in table III. The applied voltage and frequency to the master transmitter and indicator shall be set at the following points:

a. 110V ac 380 cps
b. 110V ac 420 cps
c. 120V ac 380 cps
d. 120V ac 420 cps.

When the applied voltage and frequency are changed as specified, the change in combined counter tolerances shall not exceed those specified in tables II and III.

4.6.10 LOW TEMPERATURE.- The indicator, not energized, shall be subjected to a temperature of -55° ±2° C for a period of 4 hours. While still at this temperature, the test transmitter shall be set at zero ppm. Power shall be applied to the indicator. At the end of a 10-minute period, the master transmitter shall be set to the full-scale test point. Within 180 seconds, the indicator shall read within the double scale error room temperature tolerances listed in table I. The indicator shall then meet the low temperature scale error and friction tolerances specified in table I and the counter error tolerances listed in table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:55Z
Check the source to verify that this is the current version before use.

4.6.11 HIGH TEMPERATURE.- The indicator, while energized, shall be subjected to a temperature of 71° ±2° C for a period of 4 hours and tested at this temperature for scale errors. The scale error and friction tolerances for both pointer and counter shall not exceed the tolerances specified in tables I and II.

4.6.12 POSITION ERROR.- The scale error and counter error tests shall be repeated at the test point shown in table III with the instrument turned 90° right, 90° left, face up, and face down. The change in scale error and friction shall not exceed the tolerances specified in table III.

4.6.13 FOGGING.- The indicator shall be energized and kept in a 70° ±2°C controlled ambient temperature for a minimum of 1 hour. After this period and while at this temperature, an ice cube shall be rubbed on the indicator glass face for a period of 1 to 2 minutes. The glass shall then be wiped dry (compressed air shall not be used) and the indicator inspected for evidence of water and oil fog. Evidence of fogging shall be cause for rejection. This test may be accomplished in conjunction with the high temperature scale error and friction tests.

4.6.14 LOW TEMPERATURE EXPOSURE.- The low temperature exposure test shall be conducted in accordance with procedure II of MIL-E-5272. Upon completion of the 24-hour period at -54° C and while still at this temperature, the indicator shall be subjected to the scale error and counter error tests and shall meet the tolerances specified in 4.6.10. The indicator shall then be maintained at room temperature for a period of not less than 4 hours and subjected to the scale error at room temperature test. The scale errors observed shall not exceed the tolerances specified. The indicator shall be examined for deterioration or damage due to extreme temperature exposure.

4.6.15 HIGH TEMPERATURE EXPOSURE.- The high temperature exposure test shall be conducted in accordance with procedure II of MIL-E-5272. After the 48-hour period at 71° C and while at this temperature, the indicator shall be tested for scale and counter errors. The errors shall not exceed those specified in 4.6.11. The indicator shall then be maintained at room temperature for a period of not less than 4 hours after which it shall be subjected to the room temperature scale error and counter error tests specified herein. The errors observed for pointer and counter shall not exceed the tolerances specified. The indicator shall be examined for deterioration or damage due to the extreme high temperature exposure.

GPO 828431-2

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:55Z
Check the source to verify that this is the current version before use.

4.6.16 SALT SPRAY.- The indicator shall be subjected to a salt spray test in accordance with procedure I of MIL-E-5272. The scale error at room temperature test shall be conducted immediately before and after this test. The scale error tolerances after the salt spray test shall not exceed the tolerances obtained before the test. Subsequent indicator operation shall not be affected by this test, and excessive corrosion of metal parts, including electrical receptacles, shall be cause for rejection.

4.6.17 THERMAL SHOCK.- The indicator shall be subjected to 8 cycles of immersion in water at 85° ±2° C and at 5° ±2° C as follows: Each cycle shall consist of immersing the indicator in water at 85° C for a period of 30 minutes, within 5 seconds removing it from the bath, immersing it in the other water bath maintained at 5° C for a period of 30 minutes, and then within 5 seconds again immersing it in the water at 85° C for a period of 30 minutes. Following this test, the indicator shall be subjected to the sealing test specified in 4.6.6. No leakage shall be detected. There shall be no evidence of moisture penetration nor damage to the indicator as a result of this test.

4.6.18 ENDURANCE.- The indicator shall be subjected to 1,200 hours of operation which shall include 100,000 applications of flow signals to produce an indication from 0 to 85 ±5 percent of the system range and back to 0 at a rate of 15 ±10 cpm. The scale error and friction test shall be conducted not less than 1 hour after the completion of the endurance run. The errors shall not exceed the specified tolerances.

5. PREPARATION FOR DELIVERY

5.1 PRESERVATION AND PACKAGING

5.1.1 LEVELS A AND C.- Preservation shall be in accordance with method III of MIL-P-116, and packaging shall be in accordance with MIL-P-7936. Electrical connectors shall be equipped with water-vapor-proof caps and gaskets.

5.2 PACKING

5.2.1 LEVEL A.- Indicators packaged as specified in 5.1.1 shall be packed in accordance with level A packing of MIL-P-7936 except that containers conforming to PPP-B-636 are authorized as exterior containers.

5.2.2 LEVEL B.- Indicators packaged as specified in 5.1.1 shall be packed in accordance with level B packing of MIL-P-7936 except that containers conforming to PPP-B-636 are authorized as exterior containers.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:55Z
Check the source to verify that this is the current version before use.

5.2.3 LEVEL C.- Indicators packaged as specified in 5.1.1 shall be packed in accordance with level C packing of MIL-P-7936.

6. NOTES

6.1 INTENDED USE.- The EFU-20/A indicator covered by this specification is intended for use in indicating the rate of flow and the amount of fuel transferred during in-flight refueling operations.

6.2 ORDERING DATA.- Procurement documents should specify the following:

a. Title, number, and date of this specification

b. If sampling plan B tests will not be conducted

c. Selection of applicable levels of packaging and packing.

6.3 DEFINITIONS

6.3.1 TAPPING.- The phrase "gently tapped" means the application of a light vibration sufficient to overcome the small amount of residual friction normally associated with good quality aircraft instruments.

6.3.2 HERMETIC SEAL.- A hermetic seal is defined as a perfectly closed airtight seal made between vitric or metallic materials, or both. A hermetic seal is not intended to include seals accomplished by gaskets or O-rings.

6.4 QUALIFICATION.- With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Aeronautical Systems Division, Attn: ASNPPC, Wright-Patterson Air Force Base, Ohio, and information pertaining to qualification of products may be obtained from that activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:55Z
Check the source to verify that this is the current version before use.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacturer, use, or sell any patented invention that may in any way be related thereto.

PAS/brw
ASNPE

GPO 828431-1

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:55Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE. This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12502 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commander
Research & Technology Div. and
Aeronautical Systems Div.
Systems Engineering Group (AFSC,SEP)
Wright-Patterson AFB, OH 45433

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T11:55Z
Check the source to verify that this is the current version before use.

5.8 Marking.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil -- Downloaded: 2026\0-21T11:55Z
Check the source to verify that this is the current version before use.